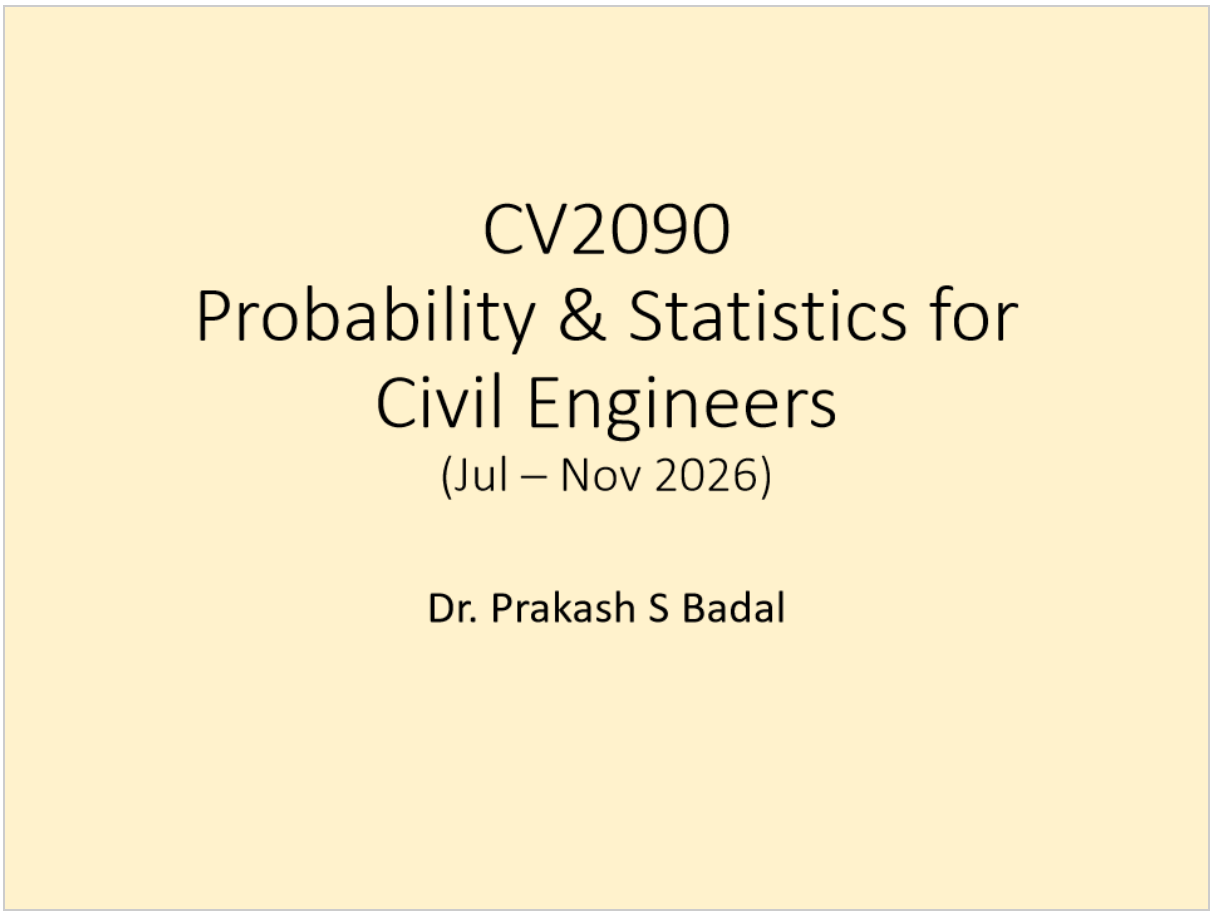
inClass-L38

Tuesday, October 27, 2026 12:20 PM

inClass-L38



L-38 (3/4; risk-n-rel)
Oct 27, 2026

Announcement

- Remaining classes:
  - Thursday, 29th Oct: Summary lecture-1
  - Monday, 2nd Nov: No class
  - Tuesday, 3rd Nov (final lecture):
    - Summary lecture-2
    - Discussion on end-semester pattern

Earlier we saw

- Space Shuttle Challenger disaster
  - → CATCH-ALL approach does NOT
  - → If everything is a priority, then nothing is.

- Typical probabilities
  - → 10^{-3} : Immediate action
  - → 10^{-6} : Nothing much can be done about it.
  - → Building $\simeq 10^{-4}$ for lifetime
  - → NPP < meltdown 10^{-5}, leakage 10^{-6}
  - → Drawing an Ace of spades 2×10^{-2}

Earlier we saw

- Limit state function
  - → Boundary between 'desirable' ($R > S$) and 'undesirable' ($R < S$) regions.
  - → $g(R,S) := R - S = 0$
  - → $$p_f = \int_0^\infty f_R(r)\,dr \int_r^\infty f_{S|R}(s|r)\,ds$$

If R & S are S.I.

$$= \int_0^\infty \int_r^\infty f_R(r)\, f_S(s)\,ds$$

$$p_f = \int_0^\infty f_S(s)\, F_R(s)\,ds$$

$$= \int_0^\infty f_R(r) \cdot [1 - F_S(r)]\,dr$$

Safety margin and Reliability index

- Safety margin

$g(R,S) := R - S = 0$ (Limit state function)

$Z = R - S$

Assumption: R & S are jointly Normal

$\mu_Z = \mu_R - \mu_S \quad ; \quad \sigma_Z^2 = \sigma_R^2 + \sigma_S^2 - 2\rho_{RS}\sigma_R\sigma_S$

$P_f = Pr(R<S) = Pr(Z<0)$

$= \Phi\left[\frac{0-(\mu_R-\mu_S)}{\sigma_Z}\right]$

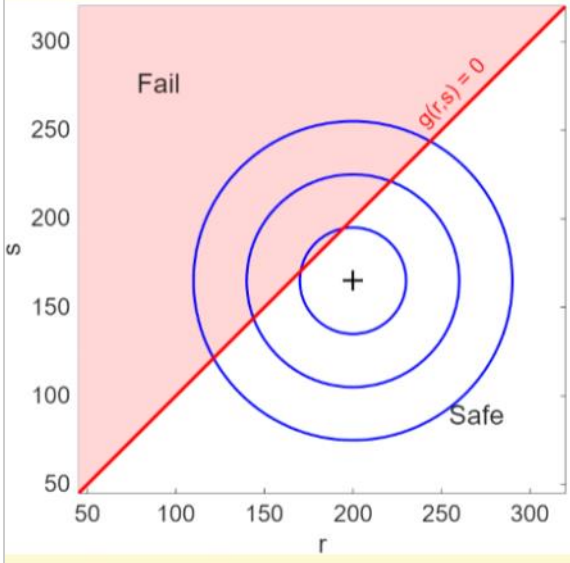
Defined as

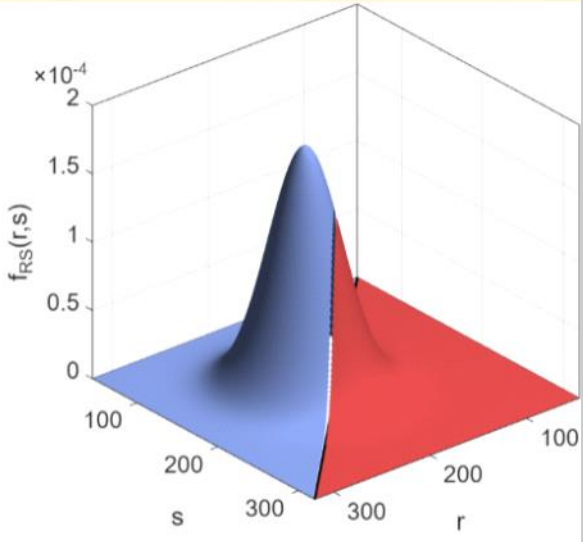
$\beta = \frac{\mu_Z}{\sigma_Z}$

Reliability Index

$P_f = \Phi[-\beta]$

Matlab demo





The R-S Problem: Safety Margin

$$g(R,S) = R - S$$

Safety margin and Reliability index

- Special case: R and S are jointly normal (ok if small c.o.v.)

The R-S Problem: Safety Factor

$$g(R,S) = \frac{R}{S} - 1$$

Safety margin and Reliability index

- Formulation in terms of (logarithmic) safety factor

$g(R,S) := \frac{R}{S} - 1$

Define $Z = \ln R - \ln S$

$\mu_Z \cong \ln \mu_R - \ln \mu_S - \frac{1}{2}(\delta_R^2 - \delta_S^2)$

$\sigma_Z^2 \cong \delta_R^2 - 2\rho_{RS}\delta_R\delta_S + \delta_S^2$

$$\beta \cong \frac{\ln r - \frac{1}{2}(\delta_R^2 - \delta_S^2)}{\sqrt{(\delta_R^2 - 2\rho_{RS}\delta_R\delta_S + \delta_S^2)}}$$

Where $r = \mu_R/\mu_S$

→ It depends only on four non-dim. params: $r, \delta_R, \delta_S, \rho_{RS}$

↑ Reliability Index

$\mathrm{E}[Y] \cong g(\mu_X) + \frac{1}{2}\left(\frac{\partial^2 g}{\partial x^2}\right)_{\mu_X} \sigma_X^2$ Second-order mean approximation

$\mathrm{Var}[Y] = \mathrm{E}[(Y-\mu_Y)^2] \cong \left[\left(\frac{\partial g}{\partial x}\right)_{\mu_X}\right]^2 \sigma_X^2$ First-order var. approx.

The R-S reliability problem

Formulation	Reliability Index β	Remark
Safety margin $Z = R - S$	$\beta = \mu_z/\sigma_z$	General definition
R and S are jointly normal	$\beta = \dfrac{r-1}{\sqrt{r^2\delta_R^2 - 2r\rho_{RS}\delta_R\delta_S + \delta_S^2}}$	Exact
Safety factor $Z = \ln R - \ln S$	SO approx. $\beta \cong \dfrac{\ln r - \frac{1}{2}(\delta_R^2 - \delta_S^2)}{\sqrt{\delta_R^2 - 2\rho_{RS}\delta_R\delta_S + \delta_S^2}}$	Approximation but general
R and S are jointly lognormal	$\beta = \dfrac{\ln\left(r\sqrt{\dfrac{1+\delta_S^2}{1+\delta_R^2}}\right)}{\sqrt{\ln\left[\dfrac{(1+\delta_R^2)(1+\delta_S^2)}{(1+\rho_{RS}\delta_R\delta_S)^2}\right]}}$	Exact but special

$p_f \cong \Phi(-\beta)$. Every row requires only four number $r, \delta_R, \delta_S, \rho_{RS}$.
Every row assumes the shape of the distribution.

Monte Carlo needs neither.

12

Thank you!